Chapter 4

TRIGONOMETRY

Learning Objectives

After studying this chapter, the students will be able to understand

- trigonometric ratio of angles
- addition formulae, multiple and sub-multiple angles
- transformation of sums into products and vice versa
- basic concepts of inverse trigonometric functions
- properties of inverse trigonometric functions



Brahmagupta

Introduction

The word trigonometry is derived from the Greek word 'tri' (meaning three), 'gon' (meaning sides) and 'metron' (meaning measure). In fact, trigonometry is the study of relationships between the sides and angles of a triangle. Around second century A.D. George Rheticus was the first to define the trigonometric functions in terms of right angles. The study of trigonometry was first started in India. The ancient Indian Mathematician, Aryabhatta, Brahmagupta, Bhaskara I and Bhaskara II obtained important results. Bhaskara I gave formulae to find the values of sine functions for angles more than 90 degrees.The earliest applications of trigonometry were in the fields of navigation, surveying and astronomy.

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DO YOU KNOW?

Currently, trigonometry is used in many areas such as electric circuits, describing the state of an atom, predicting the heights of tides in the ocean, analyzing a musical tone.

Recall

1. $\sin\theta = \dfrac{\text{side opposite to angle } \theta}{\text{Hypotenuse}}$
2. $\cos\theta = \dfrac{\text{Adjacent side to angle } \theta}{\text{Hypotenuse}}$
3. $\tan\theta = \dfrac{\text{side opposite to angle } \theta}{\text{Adjacent side to angle } \theta}$
4. $\cot\theta = \dfrac{\text{Adjacent side to angle } \theta}{\text{side opposite to angel } \theta}$
5. $\sec\theta = \dfrac{\text{Hypotenuse}}{\text{Adjacent side to angle } \theta}$
6. $\operatorname{cosec}\theta = \dfrac{\text{Hypotenuse}}{\text{side opposite to angle } \theta}$

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